Draco at 500 Light-Years

A Computational Blind Test Across 88 Constellations

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Status: Preprint — speculative paper

Preprint: Academia.edu / ResearchGate

What this is

In October 2017, the author produced two acrylic paintings derived from recurring dreams (documented by EXIF metadata: October 6 and 11, 2017).

A computational blind test compared **16 automatically-detected angular blue markers** from Painting 1 against **12 IAU constellations** projected from a grid of:
- 18 distances (10 to 2000 light-years)
- 16 galactic directions
- **3,456 combinations total**

Main result: The constellation **Draco** viewed from approximately **500 light-years** in the direction of **Cygnus** (galactic coordinates l=84°, b=2°) produces the best topological match across all tested combinations.

Metric	Value
Procrustes distance	d = 0.4521
Monte Carlo p-value	p ≈ 0 (n=10,000)
Mirror robustness	p < 0.001 in all 4 orientations
M42 lateral constraint	123.6° ✓ compatible

Repository structure

```
draco-500ly/
├── blind_test_88.py         # Main analysis script
├── analisi_dipinti_v2.py    # Secondary analysis (4 candidates)
├── data/
│   └── blue_nodes_16.csv    # The 16 detected node coordinates
├── figures/                 # Output figures
├── paintings/               # (not included — available on request)
└── README.md
```

Quick start

```
pip install astroquery astropy scipy numpy matplotlib Pillow
python blind_test_88.py
```

Place the painting JPEG files (`IMG_20171006_080030.jpg` , `IMG_20171011_081025.jpg`) in the same folder, or update the paths in the script.

The Gaia DR3 validation request

This is the most important missing piece.

We formally invite any astronomer with access to the ESA Gaia Archive to execute this query:

```
SELECT COUNT(*)
FROM gaiadr3.gaia_source
WHERE distance_gspphot BETWEEN 138 AND 168
  AND l BETWEEN 76 AND 92
  AND b BETWEEN -6 AND 10
  AND teff_gspphot BETWEEN 3700 AND 6000
```

Archive URL: https://gea.esac.esa.int/tap-server/tap

If you run this query, please open an Issue or Pull Request with your results. Any outcome — positive, negative, or inconclusive — is valuable and will be incorporated into the next paper version.

The 16 node coordinates (Painting 1)

Extracted by automatic chromatic analysis of the original JPEG, confirmed visually by the author. Normalized coordinates: (0,0) = bottom-left, (1,1) = top-right.

Node	x	y	Description
B01	0.316	0.889	top apex (head)
B02	0.532	0.779	left shoulder high
B03	0.532	0.757	left shoulder low
B04	0.445	0.694	upper left body
B05	0.664	0.626	upper right body
B06	0.487	0.618	central node
B07	0.186	0.557	left flank high
B08	0.199	0.519	left flank low

Node	x	y	Description
B09	0.681	0.434	right flank high
B10	0.699	0.399	right flank low
B11	0.588	0.390	center-right
B12	0.070	0.378	far left
B13	0.446	0.332	lower center
B14	0.334	0.268	lower left
B15	0.513	0.188	bottom center
B16	0.778	0.131	bottom right

Note for formal test: These coordinates were detected from a compressed JPEG. For a rigorous test, please scan the original painting at 600 dpi minimum and re-detect node centers with error < 1% of image dimensions.

What this result is NOT

- Not proof of reincarnation
- Not proof that dreams are prophetic
- Not proof of space travel or any supernatural phenomenon
- Not an astronomical discovery

It is an **anomalous geometric result** that deserves honest documentation and formal verification.

Citation

```
@misc{musella2026draco,
  author    = {Musella, Dmitrij},
  title     = {Draco at 500 Light-Years: A Computational Blind Test Across 88 Constellations},
  year      = {2026},
  note      = {Preprint. Academia.edu / ResearchGate},
  url       = {https://github.com/dmitrij-musella/draco-500ly}
}
```

Contact

For access to original paintings (600 dpi scan), original JPEG files with EXIF metadata, or collaboration inquiries, open an Issue in this repository.

The universe is under no obligation to make sense to us. But it is not forbidden from doing so.